(Please write your Enrollment Number) Enrollment No. 14701172022

MID-TERM EXAMINATION (Regular and Reappear)
(Course Name _B.Tech IT/AI&ML/CSE/CSE-AI) (Semester 03)
(October, 2023) OFF LINE mode

Subject Code: BIT 203	**Subject:Software Engineering**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: All questions and Compulsory. Attempt all parts of Q1 ,2 and 3	

Q1	Attempt all parts	**(2*5)**	
1.1	(a) What are the two reasons for which the software engineering is important?		
1.2	Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems:		
	(b) A system to control anti-lock braking in a car		
	(c) A virtual reality system to support software maintenance		
	(d) A university accounting system that replaces an existing system		
	(e) An interactive travel planning system that helps users plan journeys with the lowest environmental impact		
Q2	Attempt all parts	**(6,4)**	
	(a) Why is requirement engineering a critical stage of the software process? Discuss the three main activities involved in the requirements engineering process.		
	(b) Discover ambiguities or omissions in the following statement of requirements for part of a ticket-issuing system: *An automated ticket machine sells rail tickets. Users select their destination and input a credit card and a personal identification number. The rail ticket is issued and their credit card account charged. When the user presses the start button, a menu display of potential destinations is activated, along with a message to the user to select a destination and the type of ticket required. Once a destination has been selected, the ticket price is displayed and customers are asked to input their credit card. Its validity is checked and the user is then asked to input their personal identifier (PIN). When the credit transaction has been validated, the ticket is issued.*		
Q3	Attempt all parts	**(4,6)**	
	(a) Develop a sequence diagram showing the interactions involved when a student registers for a course in a university. Courses may have limited enrollment, so the registration process must include checks that places are available. Assume that the student accesses an electronic course catalog to find out about available courses.		
	(b) Using the example of a retail clothing store in a small mall, list down the relevant data flows, data stores, processes and sources/sinks. Observe several sales transaction. Draw a context diagram and level-1 diagram that represent the selling system at the store.		